MediCare — Requirements Gathering

Graduation Project — Digital Egypt Pioneers Initiative (DEPI)

Track	Full Stack .NET Web Developer
Group	ONL5_SWD5_S3
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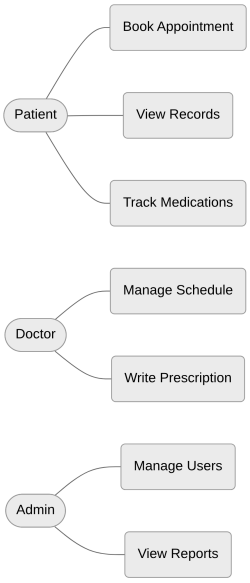
1. Stakeholder Analysis

Stakeholder	Needs
Patient	Easy booking, access to medical records, reminders
Doctor	Manage schedule, patients and prescriptions
Admin	Manage users, data and reports
Instructor	Clean code, complete documentation, working system

2. User Stories

#	User Story
1	As a **patient**, I want to search for doctors by specialization.
2	As a **patient**, I want to book, cancel or reschedule an appointment.
3	As a **patient**, I want reminders for my appointments and medicines.
4	As a **patient**, I want to see my medical records and prescriptions.
5	As a **doctor**, I want to set my working hours and vacations.
6	As a **doctor**, I want to confirm or reject appointments.
7	As a **doctor**, I want to write and print digital prescriptions.
8	As a **doctor**, I want a warning if the patient is allergic to a medicine.
9	As an **admin**, I want to approve doctors and manage users.
10	As an **admin**, I want to see reports and charts.

3. Use Cases



Use Case	Steps
Book Appointment	Choose a doctor → pick a free slot → confirm → the doctor is notified
Write Prescription	Choose a medicine → add dose and duration → allergy check → save and print
Track Medications	Receive a reminder → open the schedule → mark the dose as taken or skipped
View Reports	Open the dashboard → choose a date range → view the charts → export

4. Functional Requirements

#	Requirement
1	Register and log in with 3 roles
2	Search and filter doctors
3	Book, cancel and reschedule appointments from a calendar
4	Prevent double booking
5	Doctor dashboard for appointments and schedule
6	Medical records with file uploads
7	Digital, printable prescriptions
8	Medication catalog and allergy warning
9	Medication schedule and dose reminders
10	Real-time, SMS and email notifications
11	Admin management of doctors, patients and specializations
12	Reports with charts and PDF/Excel export

5. Non-Functional Requirements

Type	Requirement
Performance	Pages load in under 2 seconds
Security	HTTPS, hashed passwords, role-based access
Security	Protection against XSS, CSRF and SQL injection
Usability	Responsive design for mobile and desktop
Usability	Booking in 4 steps or fewer
Reliability	99% uptime and daily backups
Maintainability	NTier architecture, Repository, Unit of Work, Dependency Injection